

Lightweight Improvement Attempt

Lightweight Improvement Attempt

To investigate whether a richer feature representation could improve sentiment classification, a lightweight modification was applied to the baseline model. The original TF-IDF vectorizer extracted unigram and bigram features (`ngram_range=(1, 2)`). In the improvement experiment, trigram features were also included by changing the parameter to `ngram_range=(1, 3)`, while keeping all other settings unchanged.

The motivation behind this modification was that trigrams may capture longer sentiment expressions and additional contextual information, potentially helping the classifier distinguish the sentiment of specific aspects more accurately.

Results

Metric	Baseline	Trigram TF-IDF
Accuracy	71.43%	69.90%
Macro Precision	64.32%	61.90%
Macro Recall	59.64%	58.85%
Macro F1-score	61.31%	60.02%

The modified model produced slightly lower performance across all evaluation metrics. Although additional contextual information was introduced through trigram features, the larger feature space also increased sparsity. Since the dataset is relatively small, many trigram features occurred infrequently and therefore contributed limited useful information during training.

This experiment demonstrates that increasing feature complexity does not always improve model performance. Effective feature engineering should balance richer representations with the amount of available training data.

Simple Model Comparison

Simple Model Comparison

To compare different text classification algorithms, the baseline Logistic Regression classifier was replaced with a Multinomial Naive Bayes classifier. Both models used the same dataset, the same train-test split, and the same TF-IDF feature representation (`ngram_range=(1, 2)`), ensuring a fair comparison.

Results

Metric	Logistic Regression	Multinomial Naive Bayes
Accuracy	71.43%	67.13%
Macro Precision	64.32%	67.66%
Macro Recall	59.64%	46.32%
Macro F1-score	61.31%	47.26%

Although Multinomial Naive Bayes achieved slightly higher macro precision, its recall and Macro F1-score were considerably lower than those of Logistic Regression. The classification report showed that Naive Bayes strongly favored the Positive class, achieving a recall of **97%** for positive samples while performing poorly on the Negative and Neutral classes.

This behavior suggests that the Naive Bayes classifier was heavily influenced by the class imbalance in the dataset, where Positive examples represent the majority of training samples. Logistic Regression handled this imbalance more effectively, especially because it was trained using balanced class weights.

Overall, Logistic Regression provided the best balance across all sentiment classes and remained the stronger baseline model for this Aspect-Based Sentiment Analysis task.